PROJECT PROPOSAL

IIT 271-2

Group No: 07

Online Boarding
Finder System

(StayFinder)



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1).Introduction

1.1Project Title

Online Boarding Finder System(StayFinder)

1.2Project Background

Boarding places plays a significant role in the lives of many people, especially University Students who need to find accommodation. In today's fast-paced world, where mobility is common and opportunities are scattered, having a reliable system to locate boarding options can be invaluable.

Consider the plight of students moving to new cities or areas for their Higher Education, they often face challenges in finding suitable accommodation that meets their needs and preferences. Without proper guidance or information, they may end up in substandard living conditions or struggle to locate housing altogether. This can lead to stress, anxiety, and a host of other issues, affecting their overall well-being and productivity.

Moreover, the process of finding boarding can be time-consuming and inefficient, involving numerous phone calls, emails, and visits to various locations. This not only adds to the individual's stress but also wastes valuable time that could be spent on more productive activities.

By developing a StayFinder, we aim to address these challenges and provide a solution that simplifies the process of locating boarding options. Our system will leverage technology to aggregate and present information about available boarding facilities in a user-friendly manner. Users will be able to specify their preferences and requirements, such as location, amenities, and budget, and receive tailored recommendations based on their criteria.

StayFinder will not only benefit individuals seeking accommodation but also boarding providers, who will gain increased visibility and access to potential tenants. By facilitating efficient matchmaking between renters and landlords, our system will contribute to creating a more seamless and hassle-free boarding experience for everyone involved.

StayFinder aims to revolutionize the way people find accommodation by harnessing the power of technology to streamline the process and improve outcomes for all stakeholders.

1.3 Project Objectives

- Streamline Accommodation Search: Develop a user-friendly platform to simplify the process of finding suitable boarding options for university students, reducing the time and effort required for accommodation search.

- Facilitate Informed Decision-Making: Enable students to make informed decisions by providing accurate and up-to-date information about each boarding option, including reviews and ratings from previous tenants.

- Provide Comprehensive Housing Information: Create a database of available boarding options near the university campus, including details such as rent, amenities, location, and landlord contact information, to ensure students have access to comprehensive housing information.

- Promotion of boarding places by boarding places owners.

- Students who are new to the university and have no knowledge of the city can find their hostels without any hassle.

- Support Financial Planning: Assist students in budgeting for accommodation expenses by providing transparent information on rent prices, utilities, and other associated costs.

- Real-Time Updates and Notifications: Implement real-time updates and notifications to inform students about newly available boarding options, price changes, or relevant announcements, ensuring they stay informed throughout the accommodation search process.

1.4 PROJECT AIM

Our aim is to create StayFinder, an automated system designed to solve the challenges faced by university students in finding suitable boarding accommodations. StayFinder will streamline the process, eliminating the hassles of manual searches and providing students with a user-friendly platform to discover available boarding options tailored to their preferences and requirements.

2). Project Description

After that we have discuss with the office staff and also with the top management to understand the project scope and gather data. As a result of these discussion we have found following functional requirements and non-functional requirements

2.1 Functional / Non-functional Requirements:

❖Functional Requirements

01.User Registration and Login

The system permits users to create accounts by entering their email and password.Users can log into the system using their registered email and password.The system facilitates users in creating and managing their profiles, including options to delete or update their accounts. Warning and error messages are displayed by the system when necessary during the user registration process.

02.Profile Management

Users can personalize their profiles according to the provided structure.The system allows users to update and modify their profiles as needed.Users have the option to delete their profiles if desired.

03. Boarding Search and Filtering

Users can search for available boarding options using various filters such as location, price range, and amenities.The system enables users to refine their search results based on their preferences.

03.Boarding Listing and Management

Hosts can list their available boarding spaces on the platform, including details such as location, price, and facilities.Hosts have the ability to manage their listings, including updating information and marking availability.

04.Messaging and Communication

The system provides a messaging platform for users to communicate with each other regarding boarding inquiries and arrangements.

05.University Hostel Searching

Students who are new to the university and have no knowledge of the city can find their hostels without any hassle.

06.Ratings and Reviews

Users can rate and review boarding accommodations based on their experience.The system displays ratings and reviews to help users make informed decisions when choosing a boarding option.

❖ Non-functional Requirements

- **Performance:** Ensure fast response times for searches, bookings, and page loading.
- **Scalability:** The system should be able to handle a large number of users and hostels without significant degradation in performance.
- **Reliability:** The system should be available and operational 24/7 with minimal downtime.
- **Security:** Implement robust security measures to protect user data, including encryption of sensitive information and secure transmission protocols.
- **Usability:** The user interface should be intuitive and easy to navigate, catering to users with varying levels of technical proficiency.
- **Compatibility:** The system should be compatible with a wide range of devices and browsers to accommodate different user preferences.
- **Accessibility:** Ensure that the platform is accessible to users with disabilities, following accessibility guidelines such as WCAG (Web Content Accessibility Guidelines).
- **Compliance:** Adhere to relevant regulations and standards governing data protection, online transactions, and accessibility.
- **Performance Monitoring:** Implement tools for monitoring system performance, usage patterns, and user feedback to continuously improve the system.
- **Localization:** Support localization efforts to adapt the system to different regions and cultures, including currency, date formats, and language preferences.
- **Supportability:** Provide documentation, training materials, and customer support resources to assist users and address any issues that may arise

2.2 User Levels, User Roles

1. **Guest**:
    - These are users who are visiting the website but have not registered.
    - They should be able to search for available boardings, view details, and contact the host for more information.
2. **Registered User**:
    - These are users who have registered on the website.
    - In addition to the features available to guests, they might also be able to save their searches, bookmark favorite boardings, and leave reviews.
3. **Host**:
    - These are users who have boardings to rent out.
    - They should be able to list their boardings, manage their listings, respond to inquiries, and receive payments.
4. **Administrator**:
    - These are users who manage the website.
    - They should be able to manage all listings, users, and reviews.
    - They might also handle disputes and have access to reporting tools.
5. **Moderator**:
    - These are users who help the administrators in managing the website.
    - They might be able to moderate reviews, manage user reports, and assist users with their queries.
6. **Support Staff**:
    - These are users who provide customer service.
    - They should be able to assist both hosts and registered users with any issues they might have.

3).METHODOLOGY

We use waterfall methodology for develop our project.

Waterfall Methodology

A traditional method for developing software that follows a sequential and linear procedure is the waterfall model. The development process in this paradigm proceeds through multiple phases, including requirements gathering, design, implementation, testing, deployment, and maintenance, in a continuous downward flow akin to a waterfall. Before going on to the next phase, each one must be done, and it can be challenging to incorporate changes once a phase is complete.When working on projects with clearly defined criteria that are unlikely to change dramatically, this strategy is frequently employed.

It can occasionally be inflexible and less flexible to changes during the development process, despite the fact that it offers a clear framework and is simple to understand.In general, the waterfall model provides an organized approach to managing software projects from inception to completion.



In this methodology, we divided our project into 6 main steps. That is,

1. Requirements Gathering:

- We define the scope of the project, including features, functionalities, and user requirements for the online boarding finder system.
- Next we gather detailed requirements through meetings, interviews, and surveys with stakeholders, such as hostel owners, users, and administrators.
- In the end in this step we document all requirements in a comprehensive requirements specification document.

2. System Design:

- We create a high-level system architecture that outlines the overall structure of the online boarding finder system, including databases, servers, and interfaces.
- Next develop detailed technical specifications based on the requirements, specifying the technologies, frameworks, and tools to be used.
- Design the user interface (UI) and user experience (UX) of the system

3. Implementation:

- In the first step in this part we start coding the system according to the design specifications.
- Next we develop different modules or components of the system, such as user authentication, boardingl search, booking, and payment processing.
- conduct regular code reviews and testing to ensure adherence to coding standards and identify any defects early in the development process.

4. Testing:

- We use various types of testing, including unit testing, integration testing, and system testing, to verify that each component and the entire system functions correctly.
- Create test cases based on the requirements and execute them systematically to identify and address any bugs or issues.
- Conduct user acceptance testing (UAT) with actual users to validate that the system meets their needs and expectations.

5. Deployment And Maintenance:

In deployment Part ,

- we prepare the system for deployment to the production environment, including setting up servers, databases, and other necessary infrastructure.
- Next we deploy the system to the live environment, ensuring a smooth transition from the development environment.
- Finally in this step we monitor the system closely during the initial deployment phase to address any performance or stability issues that may arise.

In maintain part ,

- We provide ongoing support and maintenance for the online boardingl finder system, addressing any bugs, issues, or user feedback that emerge after deployment.
- Make periodic updates and enhancements to the system to keep it up-to-date with changing user needs and technological advancements.
- Finally We monitor system performance and security regularly and implement measures to optimize and protect the system as needed.

Every stage of the Waterfall approach is finished in order, with the output from one phase being used as the input for the subsequent phase. Given that the Waterfall technique stresses an organized and linear approach to development, modifications to requirements or design are typically discouraged once the project advances past the requirements collection phase.

4).Resources

4.1. Software

- **Edge Web Browser**

  Edge is a popular web browser because it will work well on different devices and operating system . it also has tools that help developers find and fix problems in their web sites.

- **XAMPP**

  XAMPP is a program that helps to build websites on our own computer. It sets up everything we need, like a web server and a database.so we can know about how our website works before making it live.

- **NetBeans**

  NetBeans is a program that helps us to write code for making softwares.it has a lot of useful features that make coding easier, like checking for mistakes and organizing our computer.

4.2 Hardware

- **Computers**

  We select laptop PCs as our hardware because of their strength and versatility. They are capable of doing challenging jobs, and we may improve them further by giving them more memory and storage. They may be tailored to meet our needs and used for both work and enjoyment.

  - **Smartphones And Tablet pc's (Android or IOS)**

4.3 Technologies

- **HTML**

  Html is used to structure and present content on the web. It provides the basic building blocks for creating the structure of a web page.

- **CSS**

CSS is used to style the html elements. It allows you to control the colors, fonts, layout etc…

- **JavaScript**
  JavaScript is used to handle user interactions, perform client-side validations and manipulate the document object model.

- **PHP**
  PHP is a server-side scripting language. It is used in conjunction with web servers like Apache, to handle form submission and interact with databases.

- **MYSQL**
  MySQL is a database management system. It provides a reliable and scalable solution for storing and retrieving data in web applications. PHP can communicate with MYSQL to perform database operations like storing user information, retrieving data for display, or executing complex queries.

5).Project Plan (Gantt Chart)

	Task	week											
		01	02	03	04	05	06	07	08	09	10	11	12
01	Identify problem domain												
02	Requirement gathering												
03	Create Project Proposal And Get Approval												
04	Proposal Submission And Presentation												
05	System Design												
06	Frontend Development												
07	Backend Development												
08	Database Design												
09	Testing												
10	Deployment												
11	User Training												
12	Create Final Report ,Submission and Presentation												
13	Documentation												

6).REFERENCES

1). 28 Free E-Book and Course for Web Developers at https://enlear.academy/28-ebook-and-course-free-for-web-developmen t-4f01a4d1f4a6 .

2). Learn PHP Tutorial at https://www.javatpoint.com/php-tutorial .

3). W3 School WebSite

- w3 School PHP Tutorial at https://www.w3schools.com/php/default.asp .
- w3 school SQL Tutorial at https://www.w3schools.com/sql/default.asp .
- w3 school JavaScript Tutorial at https://www.w3schools.com/js/default.asp .
- w3 school HTML Tutorial at https://www.w3schools.com/html/default.asp .
- w3 school CSS Tutorial at https://www.w3schools.com/css/default.asp .
- w3 school Java Tutorial at https://www.w3schools.com/java/default.asp .
- w3 school PHP Tutorial at https://www.w3schools.com/php/default.asp .
- w3 school Bootstrap Tuutorial at https://www.w3schools.com/bootstrap/bootstrap_ver.asp .
- w3 school React Tutorial at https://www.w3schools.com/react/default.asp .

4). HTML and CSS: Design and Build Websites First Edition by Jon Duckett at https://www.amazon.com/HTML-CSS-Design-Build-Websites/dp/1118008189 .

5). You Tube

- Bootstrap Tutorial on You Tube at https://www.youtube.com/playlist?list=PLK5U0tyd34tCrlOvIfeVO6YzW_z-f0jxk .
- Java Tutorial for Beginners on You Tube at https://www.youtube.com/playlist?list=PLsyeobzWxl7pe_IiTfNyr55kwJPWbgxB5 .